

# Active Diagnostic Probing for Backdoor Sample Cleansing in Contrastive Learning

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**Abstract**—Contrastive learning has emerged as an effective paradigm for learning transferable representations from large-scale unlabeled data. However, its reliance on uncurated pre-training data exposes it to data-poisoning backdoor attacks. In this work, we study backdoored sample cleansing for unlabeled contrastive-learning datasets and develop an active diagnostic probing framework. The key idea is to transform difficult-to-separate static representations into probe-induced response representations, calibrate benign responses using a small cross-domain reference set, preserve local response topology with UMAP, and identify suspicious samples with reference-anchored DBSCAN. Extensive experiments will evaluate cleansing effectiveness, post-cleansing security and utility, and computational overhead.

**Index Terms**—Contrastive learning, self-supervised learning, backdoor attacks, poisoned sample detection, data cleansing.

## I. INTRODUCTION

CONTRASTIVE learning (CL) has become a fundamental paradigm for self-supervised representation learning, enabling neural networks to learn transferable visual representations from large-scale unlabeled datasets. By encouraging different augmented views of the same instance to produce similar representations while maintaining discriminability among different instances, contrastive learning has achieved strong performance across a wide range of downstream tasks. However, the reliance on large-scale and potentially uncurated data sources introduces an important security risk: an adversary may inject a small number of poisoned samples into the unlabeled pre-training dataset and implant a hidden backdoor into the learned encoder.

In this work, we focus on preventive sample cleansing before retraining a clean contrastive encoder. Our central idea is to move from passive feature inspection to active diagnostic probing. A carefully designed probe is applied to suspicious samples, and the resulting representation changes are characterized as response features. These response features are then calibrated with a small trusted benign reference set and analyzed through local topology and density.

## II. RELATED WORK

### A. Backdoor Attacks on Contrastive Learning

Write the related work here.

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Manuscript received Month xx, 2026; revised Month xx, 2026.

### B. Backdoor Defenses for Contrastive Learning

Write the related work here.

### C. Representation-Based Anomaly Detection

Write the related work here.

## III. THREAT MODEL

### A. Attacker Capabilities

Describe the poisoning ratio, trigger, attack target, and attacker knowledge.

### B. Defender Capabilities

Describe access to the encoder, unlabeled dataset, and small trusted reference set.

### C. Assumptions and Defense Objective

Describe the trusted-reference and randomized-probe assumptions, together with the cleansing objective.

## IV. MOTIVATION AND KEY OBSERVATIONS

## V. METHODOLOGY

## VI. EXPERIMENTAL EVALUATION

## VII. DISCUSSION AND LIMITATIONS

## VIII. CONCLUSION